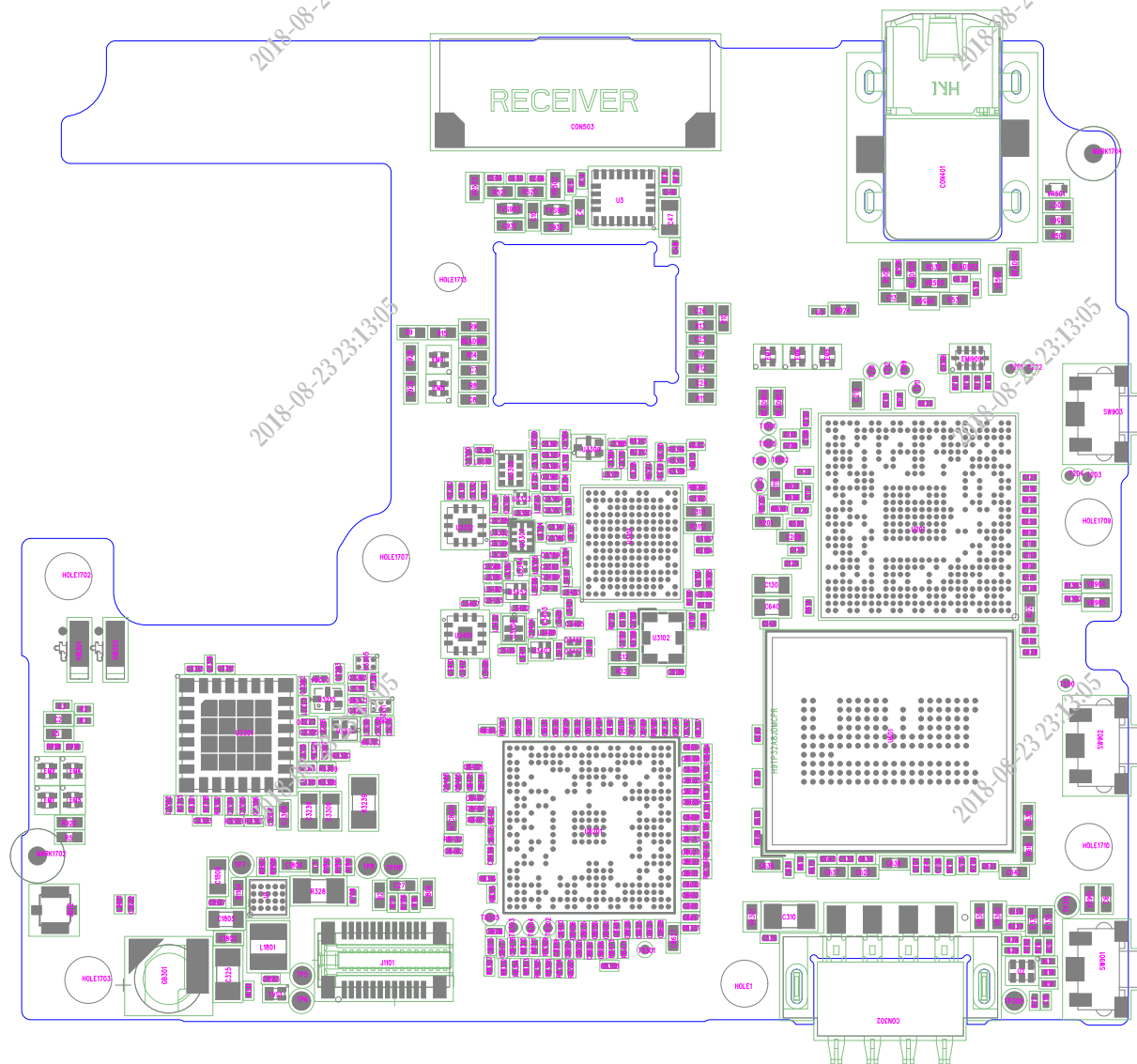
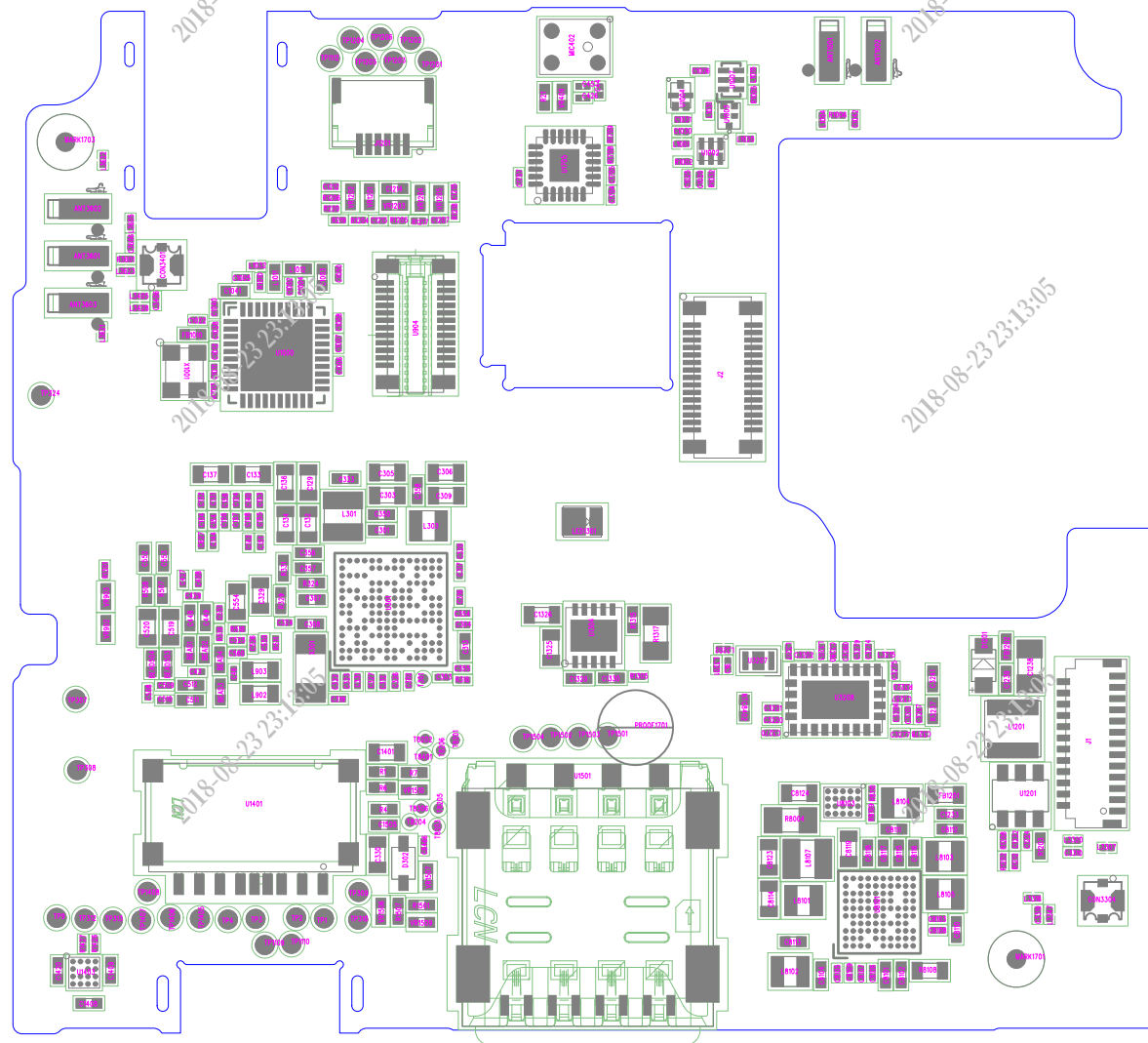
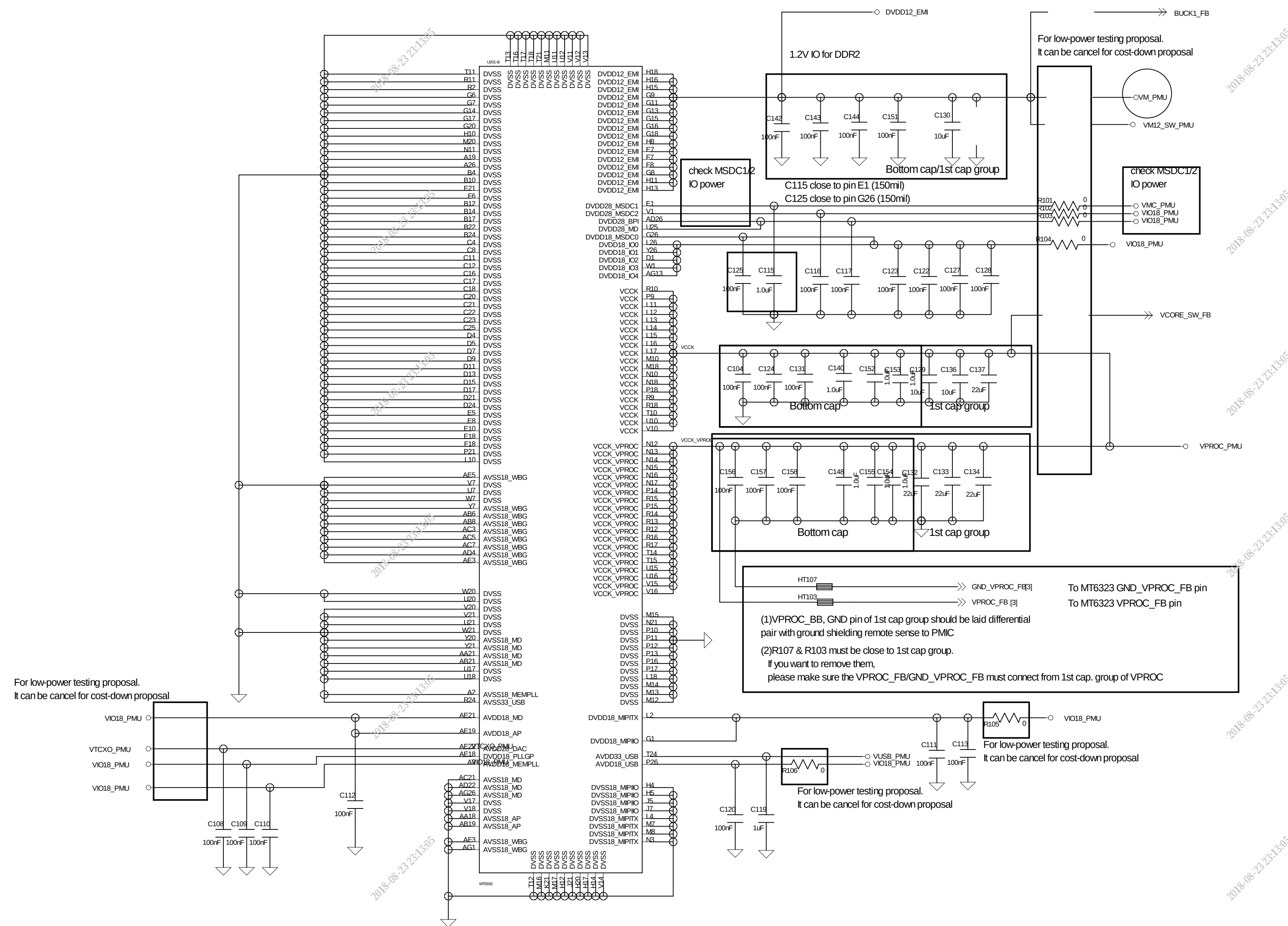








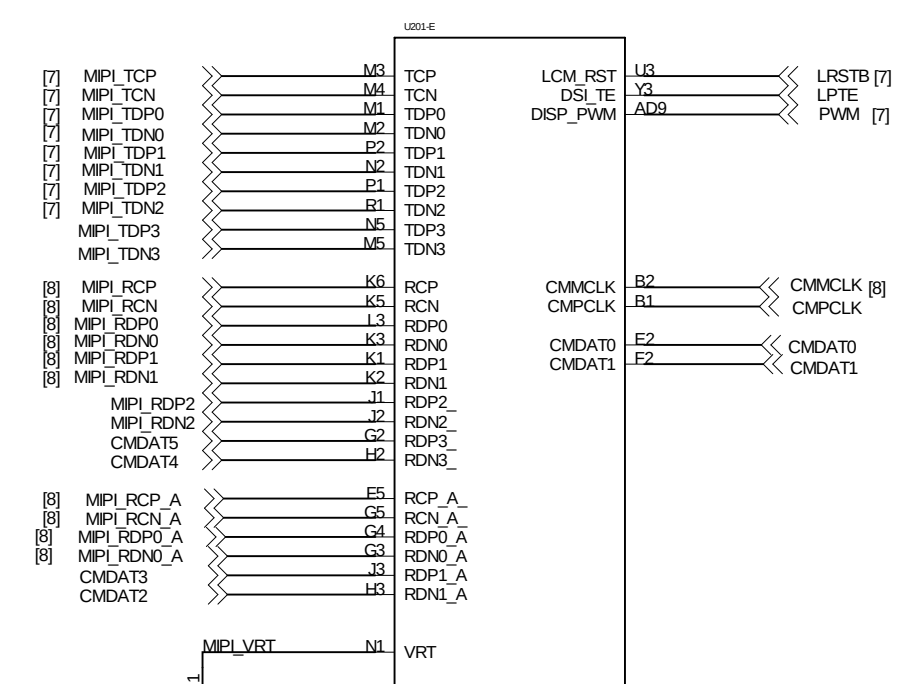
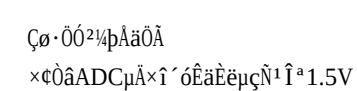
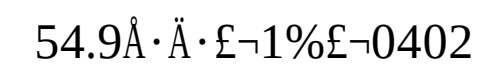
MT6582 R119=NF, R101=0, R102=NF, R120=NF, R110=0

MT6592 R119=0, R101=NF, R102=0, R120=0, R110=NF

MT6582 C135, C137, 22uF

MT6592 C135=22uF, C137=22uF

TITLE:	<TITLE>	REV:	<REV>
DOCUMENT NO:	.01_MT6582_POWER	SIZED:	A1
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	<DESIGNER>	Last Saved Date:	2014-7-3
SHEET:	1	OF	25



Close to MT6582

For earphone & MSDC hotplug EINT,
please choose EINT[0:15]
with HW de-bounce function.

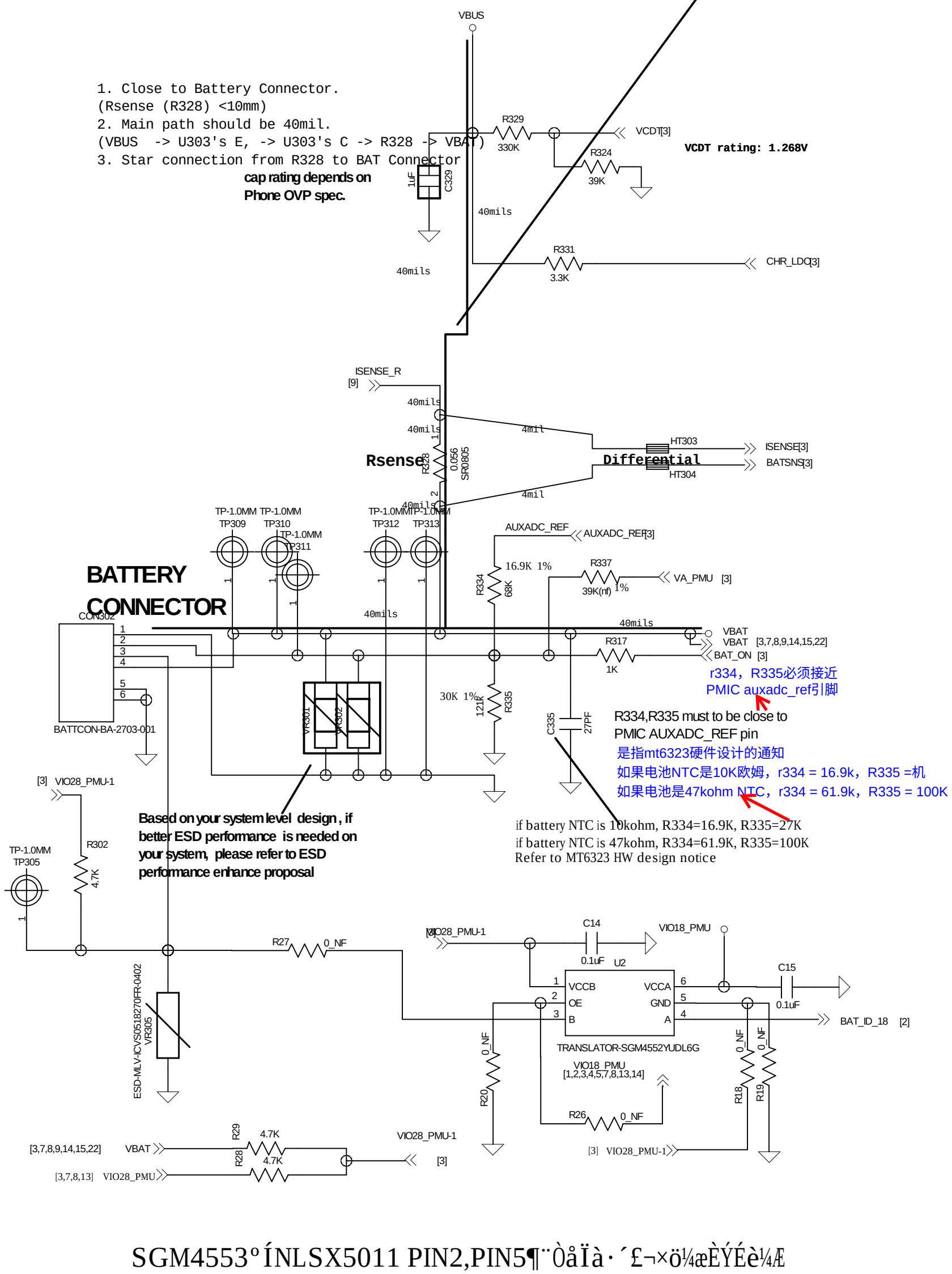
Notice :
Please choose EINT[0:15] with
HW debounce for
mechanism plug in/out related application.
Ex: earphone, MSDC, SIM hot-plug

TITLE:	<TITLE>		REV:	<REV>	
DOCUMENT NO.:	02_MT6582_BASEBAND		SIZED:	A1	
DEPARTMENT:	Hardware DEPT.				
COMPANY:					
DESIGNER	<DESIGNER>	List Saved Date:	2014-7-3	SHEET:	2 OF 25

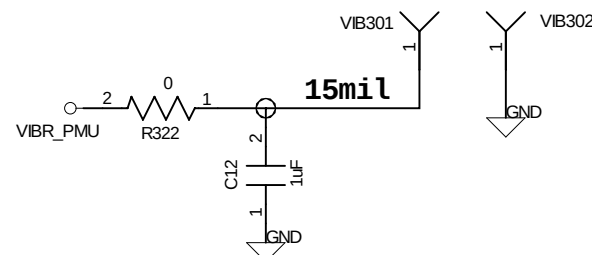
Before you select BJT , please take power dissipation into consideration.
Refer to MT6323 design notice

Charger

1. Close to Battery Connector.
(Rsense (R328) <10mm)
2. Main path should be 40mil.
3. Star connection from R328 to BAT connector
cap rating depends on
Phone OVP spec.



VIBRATOR



ISENSE/BTSNS 4m1 differential to Rsense 差分检测电阻

close to PMIC

Add Zener Diode 500mV
Place on the path from VBAT to IC
(Battery connector or test point or IO connector)
VF : 4.89V~5.38V

Refer to MT6323 design notice for Zener selection

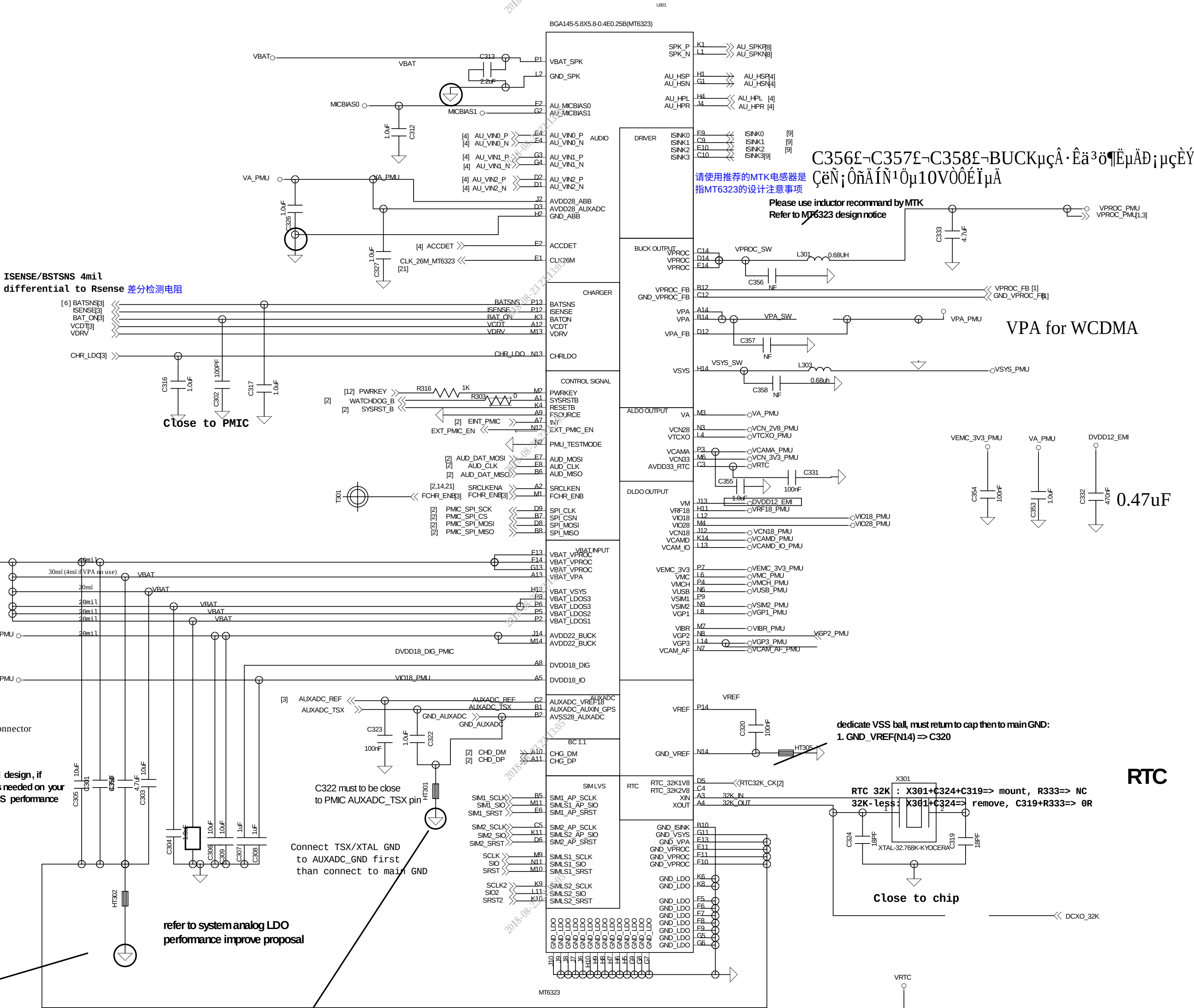
Based on your system level design, if better EOS performance is needed on your system, please refer to EOS performance enhance proposal

Connect TSX/XTAL GND to AUXADC_GND first than connect to main GND

refer to system analog LDO performance improve proposal

Refer to MT6323 design notice for Buck GND layout rule

Refer to GPS co-clock layout rule



C356E-C357E-C358E-BUCK 0.47uF
Please use inductor recommend by MTK
Refer to MT6323 design notice

VPA for WCDMA

0.47uF

dedicate VSS ball, must return to cap then to main GND:
1. GND_VREF(N14) => C320

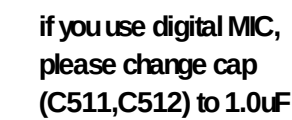
RTC

Close to chip

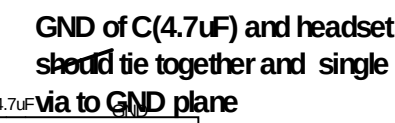
=> for longer RTC time sustain after battery remove, please refer to RTC design notice
=> for longer RTC time sustain after battery remove, please refer to RTC design notice

Symbol	LPDDR2/1.2V	PCDDR3L/1.35V	PCDDR3/1.5V	LPDDR1/1.8V	Default
SPI_CSN	H	L (20K)	H	L (20K)	PU
AUD_MOSI	L	H (20K)	H (20K)	L	PD

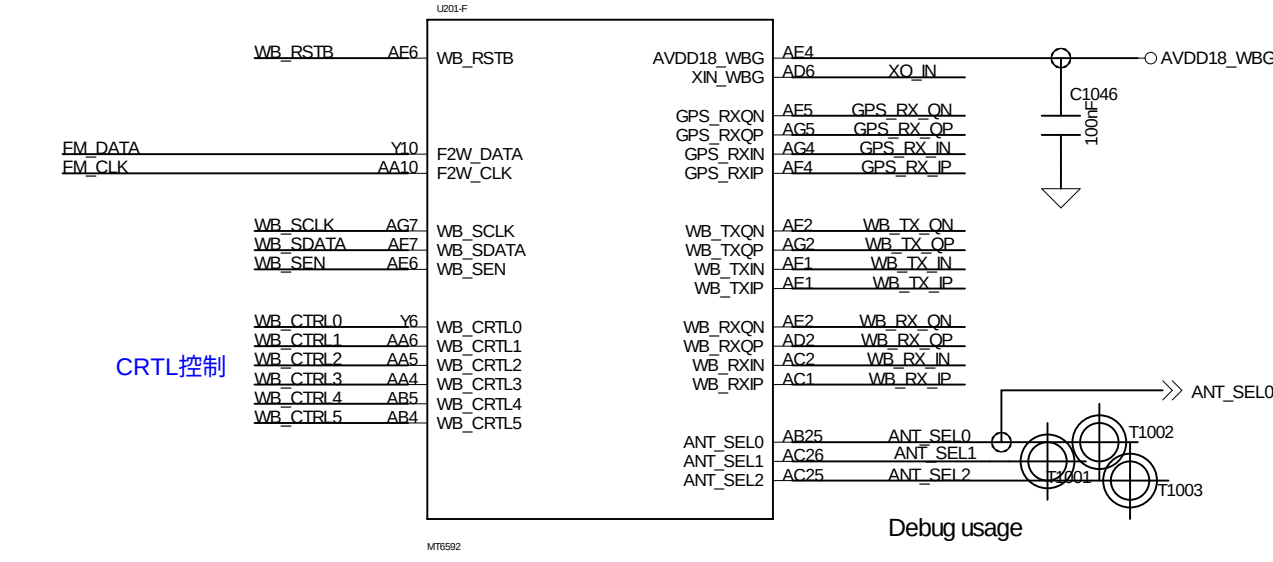
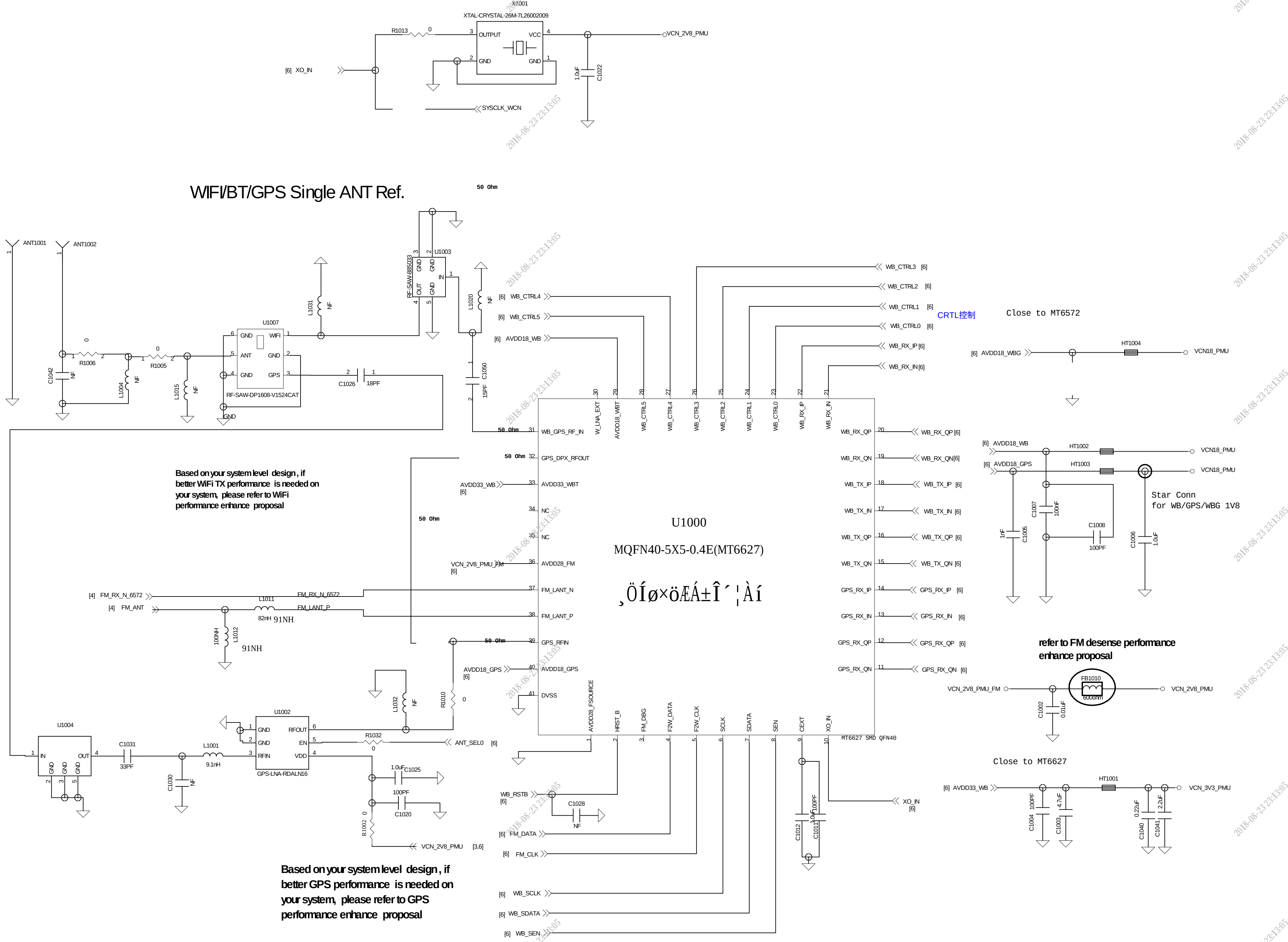
Receiver



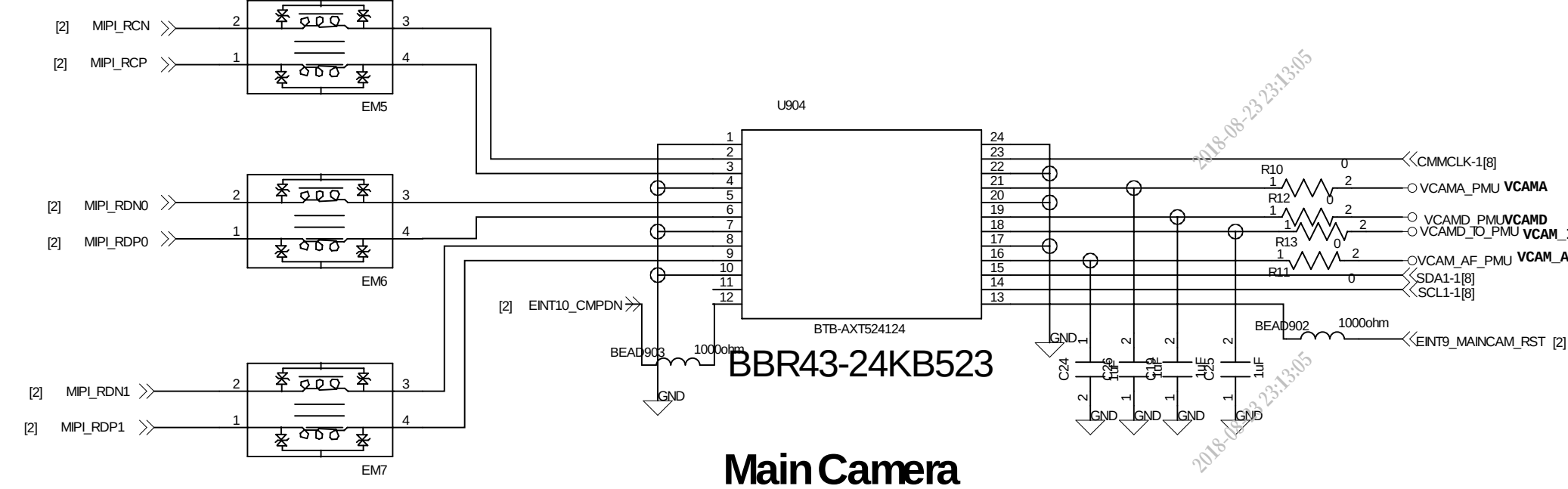
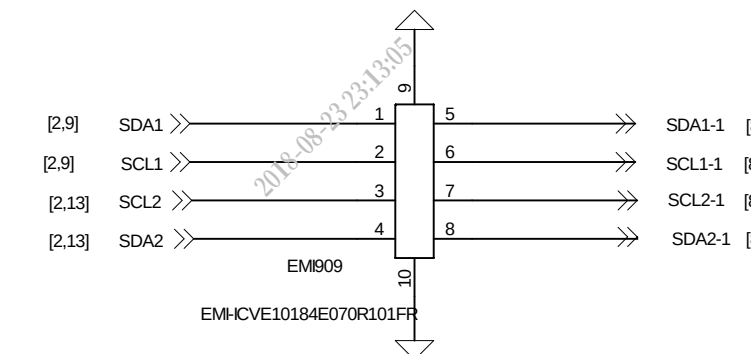
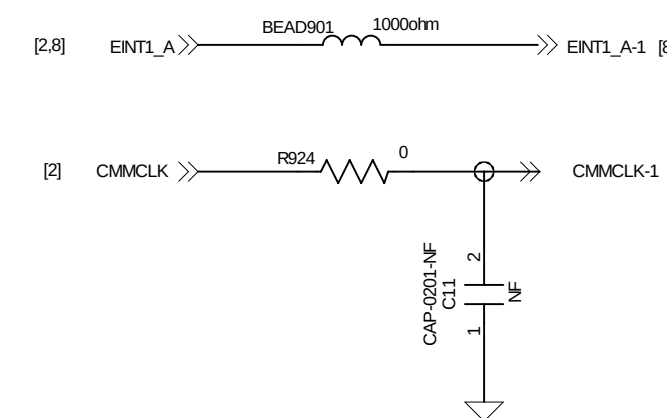
Analog MIC

[illegible]

TITLE: <TITLE>		REV: <REV>
DOCUMENT NO: 05_AUDIO	SIZED: A1	
DEPARTMENT: Hardware DEPT.		
COMPANY: 		
DESIGNER: <DESIGNER>	Last Saved Date: 2014-7-3	SHEET: 4 OF 25



FRONT_camera+LP Sensor+SPK



OCP8110 R1315=82K R1317=0.47

FLASH LED

TITLE: <TITLE>		REV: <REV>
DOCUMENT NO: 13_CAMERA	SIZED: A1	
DEPARTMENT: Hardware DEPT.		
COMPANY: 		
DESIGNER: <DESIGNER>	List Saved Date: 2014-7-3	SHEET: 8 OF 25

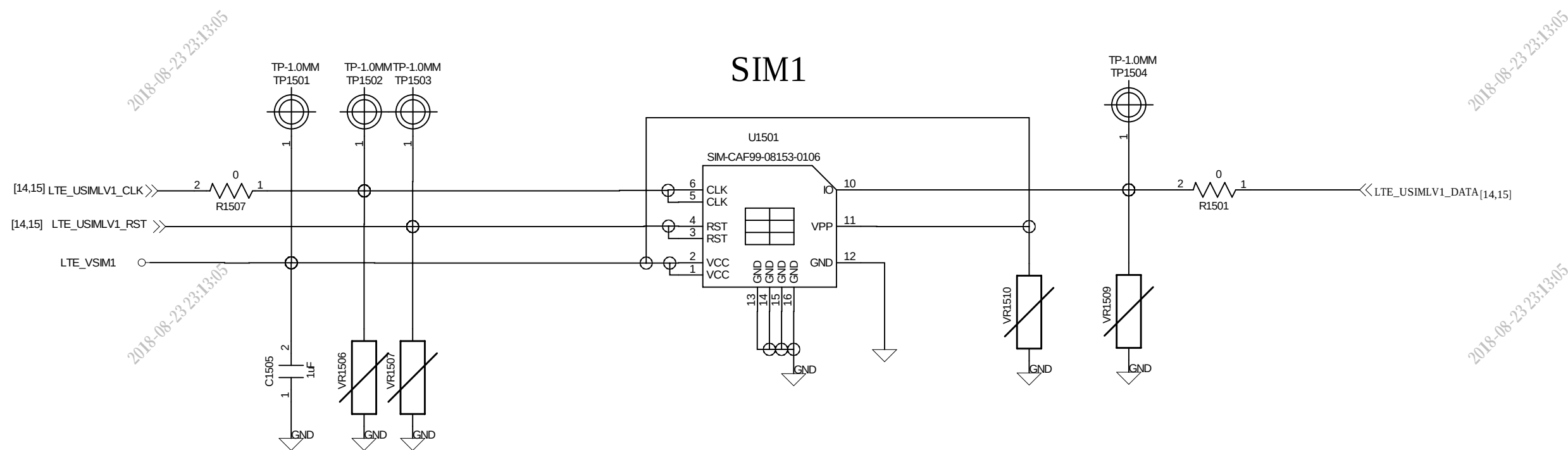


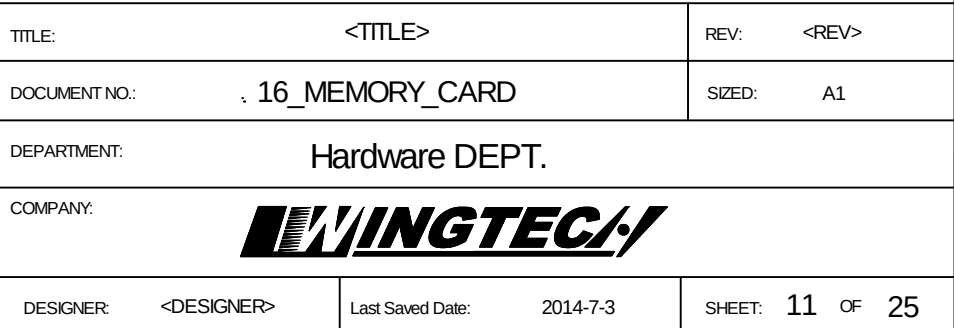
If switching charger is used:

- (1) R1801-R1805, C1801-C1806, L1801, U1801 are needed
- (2) U303, U304 change to NC
- (3) R328 change to 56m Ohm

Switching Charger

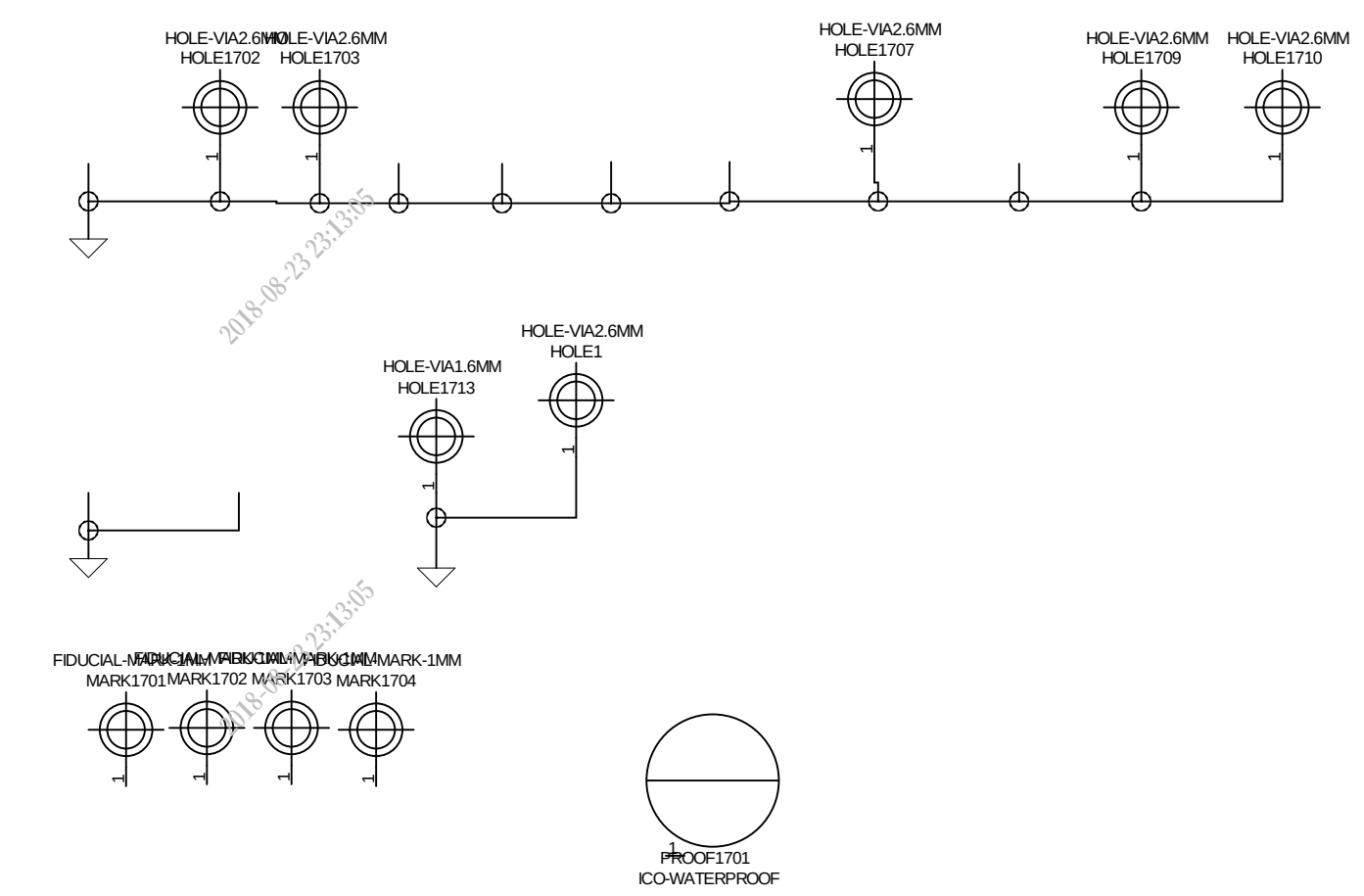
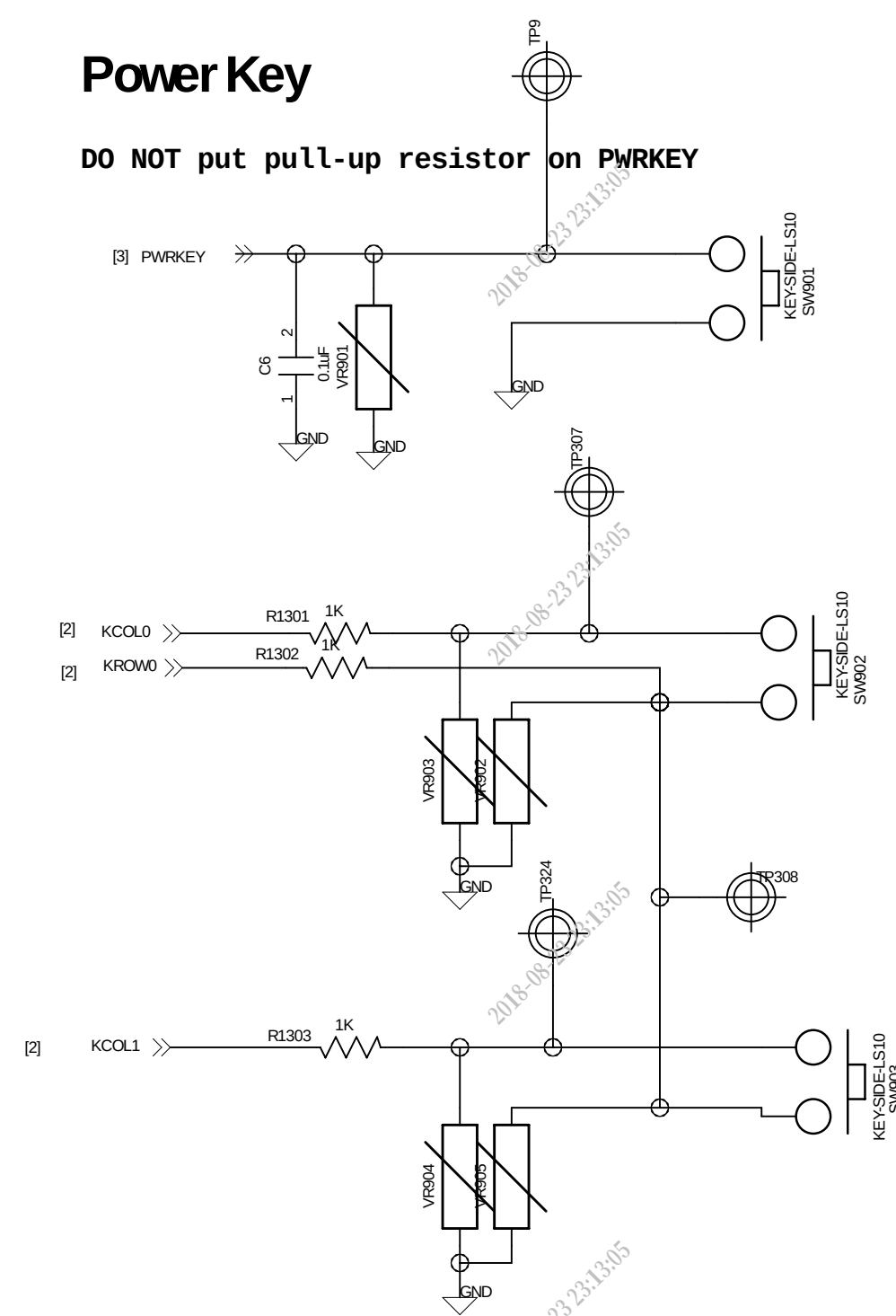
TITLE: <TITLE>		REV: <REV>	
DOCUMENT NO.: 14_USB		SIZED: A1	
DEPARTMENT: Hardware DEPT.			
COMPANY: 			
DESIGNER: <DESIGNER>	Last Saved Date: 2014-7-3		SHEET: 9 OF 25





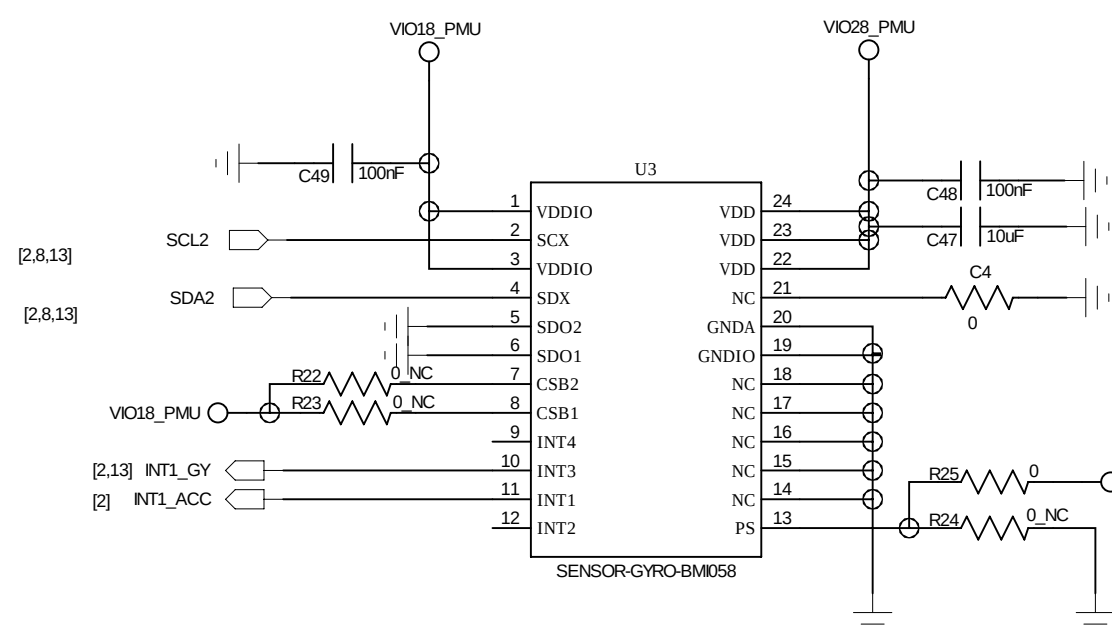
Power Key

DO NOT put pull-up resistor on PWRKEY



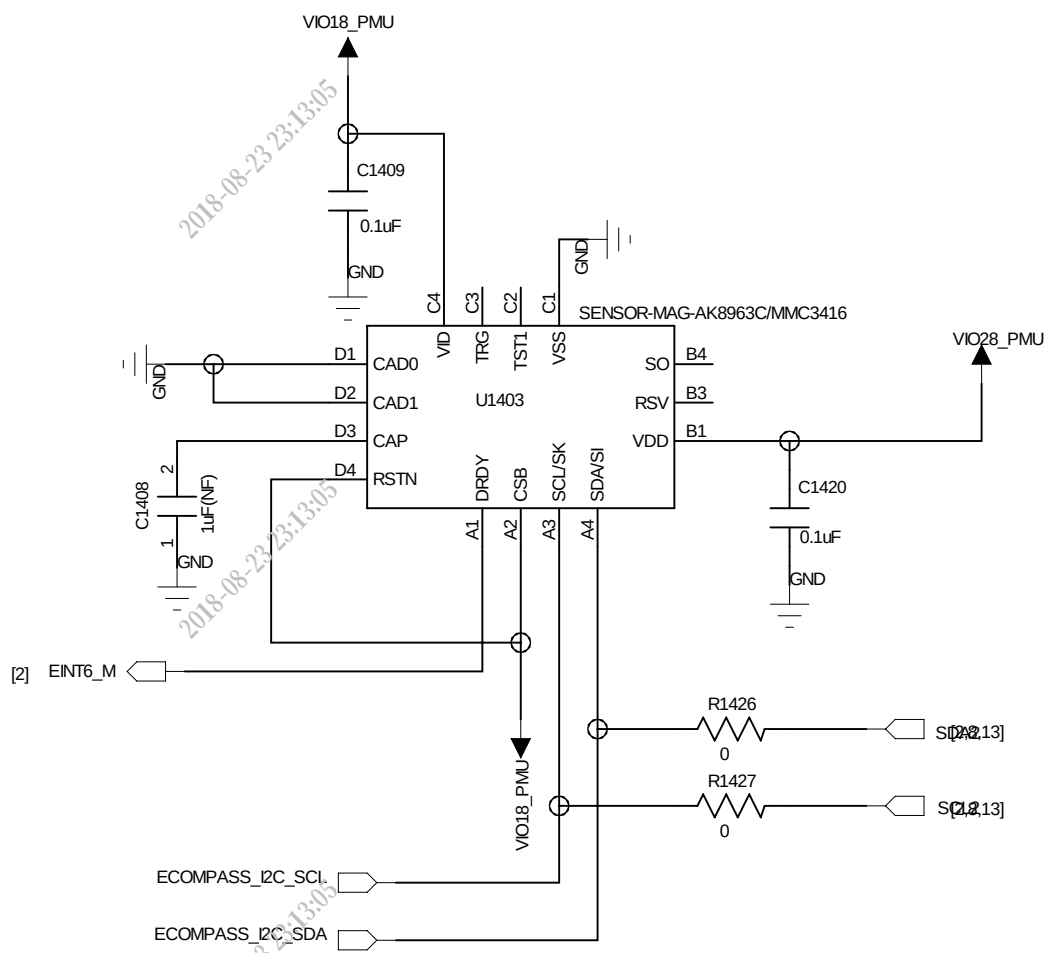
TITLE: <TITLE>		REV: <REV>
DOCUMENT NO.: 17_KEY_LEDS		SIZED: A1
DEPARTMENT: Hardware DEPT.		
COMPANY: 		
DESIGNER: <DESIGNER>	Last Saved Date: 2014-7-3	SHEET: 12 OF 25

Gyroscope



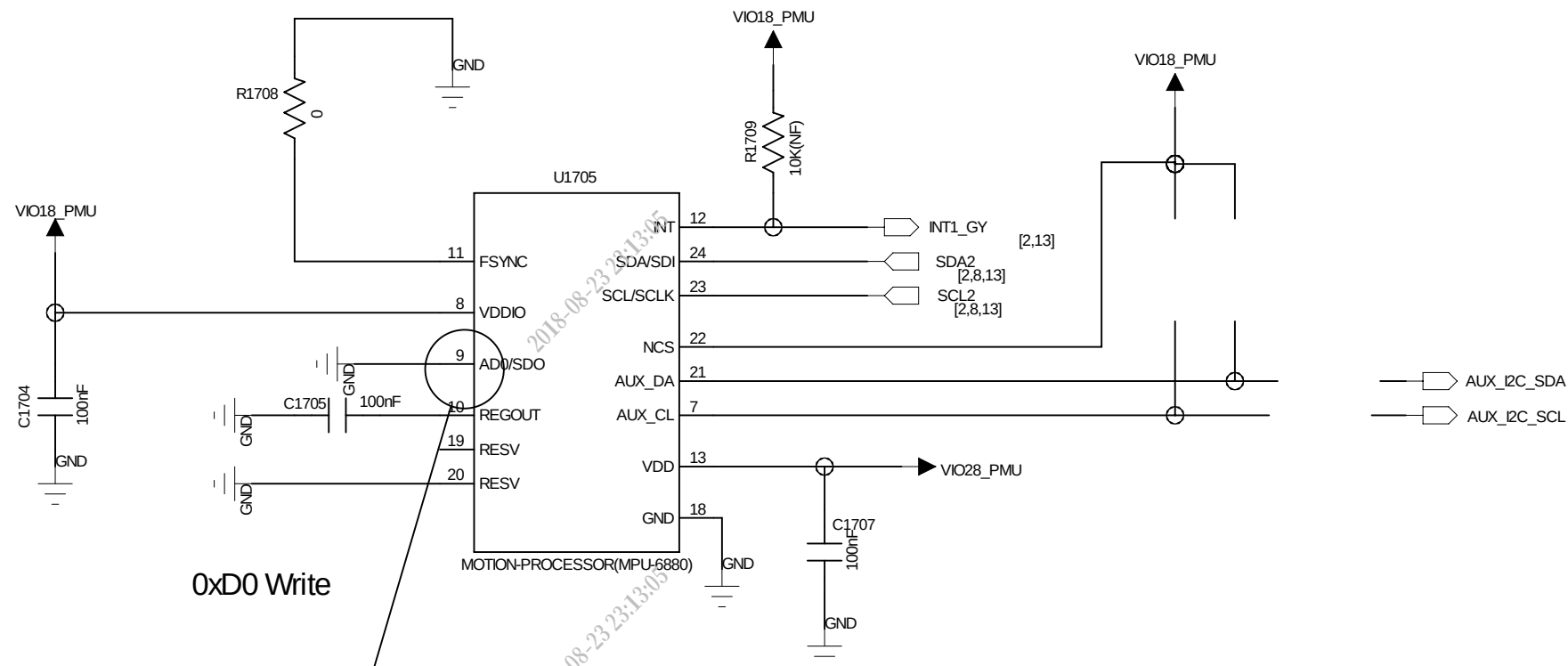
LSM330 C12=0.01uF(25V)
LSM330 R22 R23=0

M-Sensor

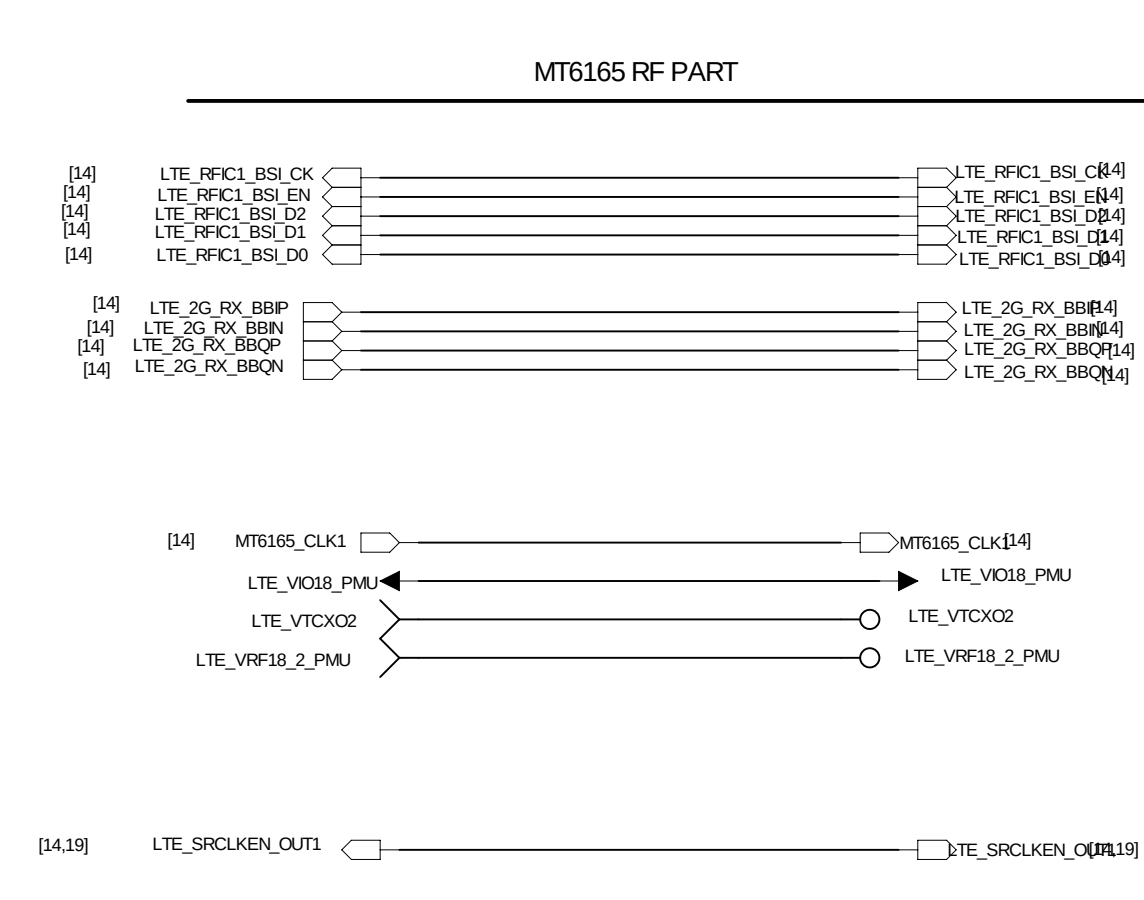
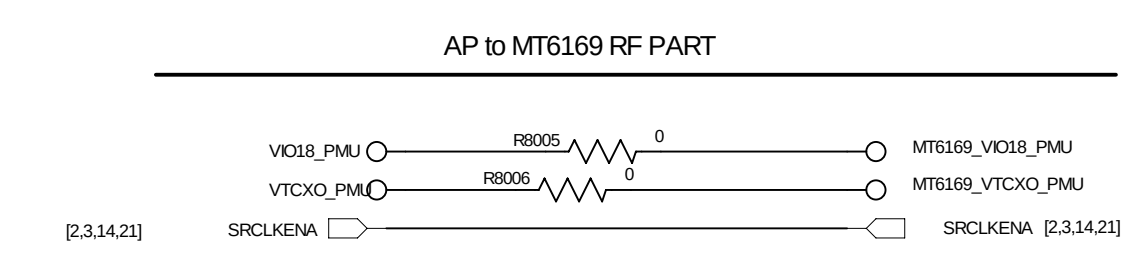
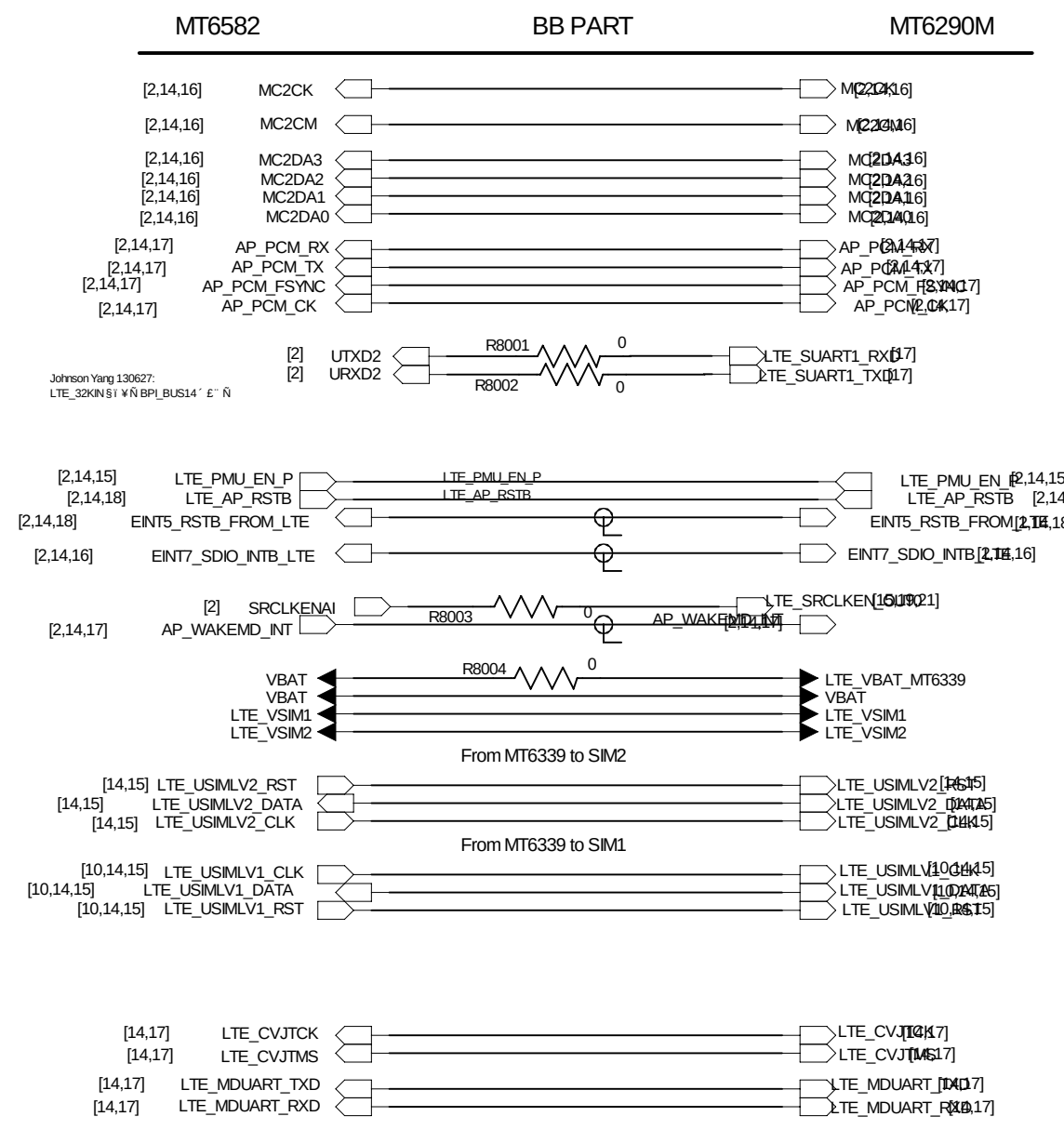


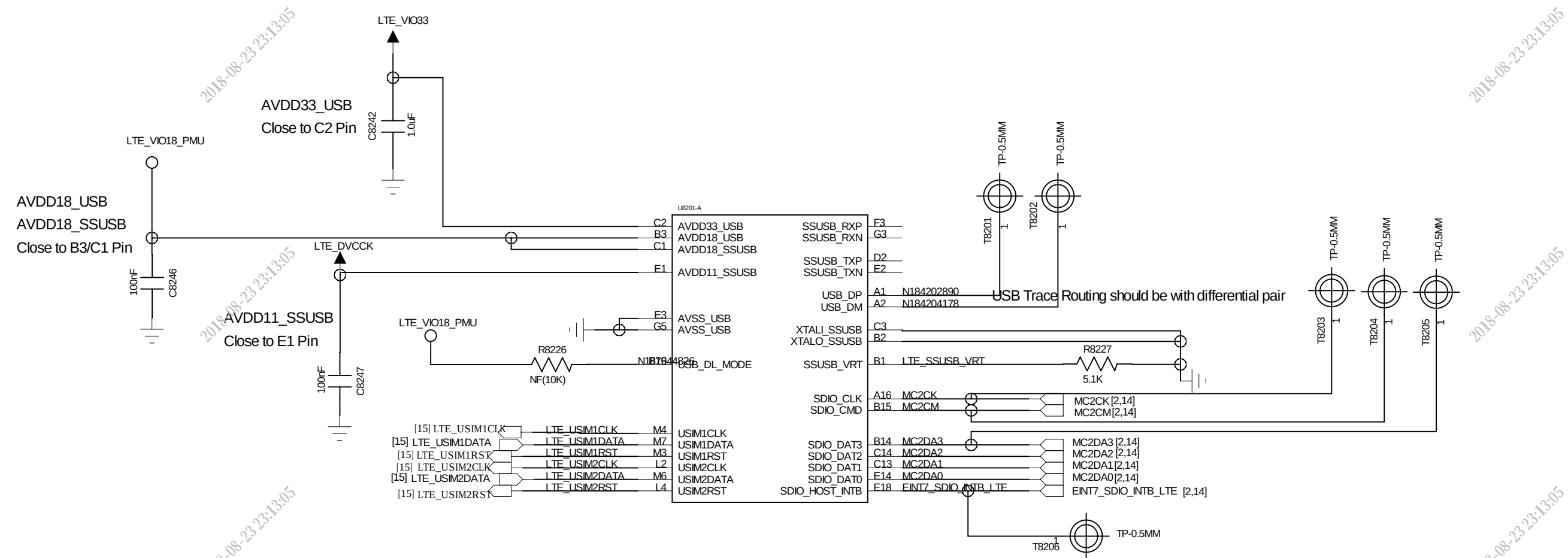
0x18 Write

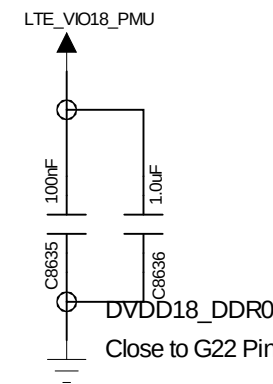
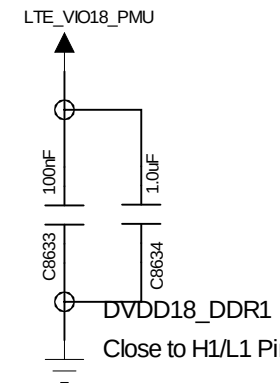
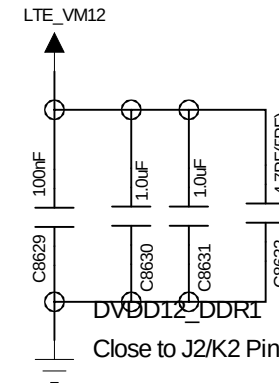
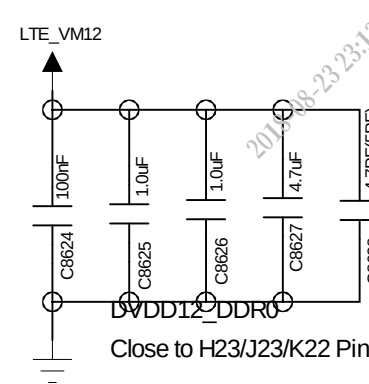
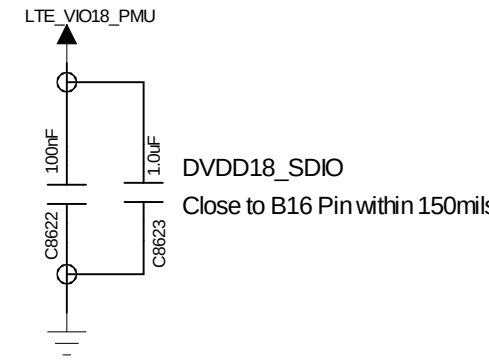
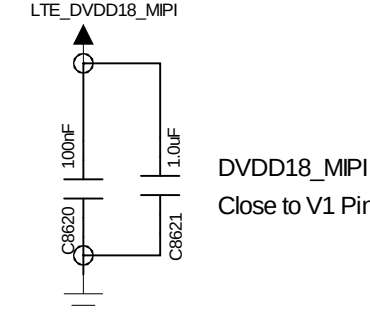
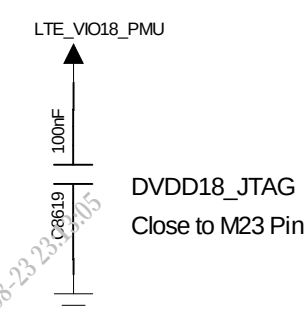
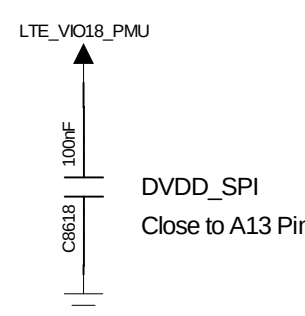
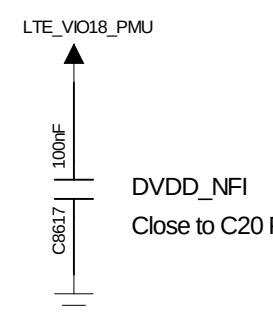
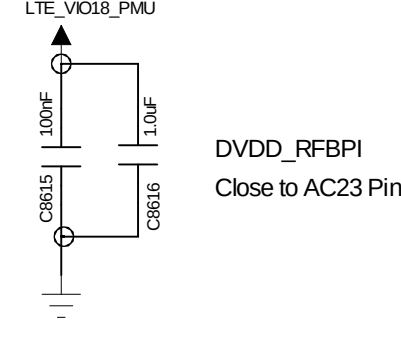
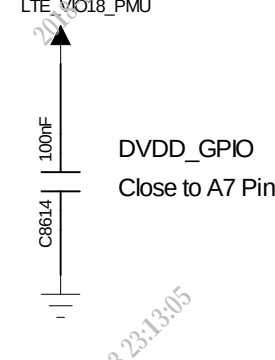
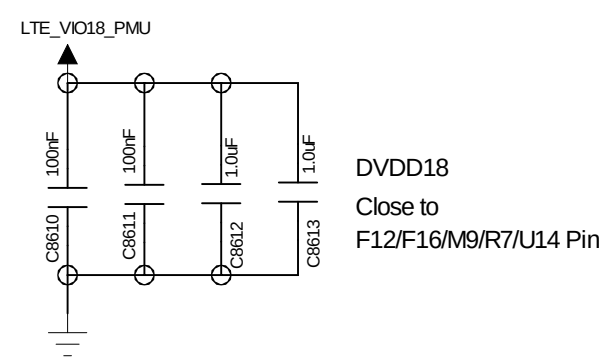
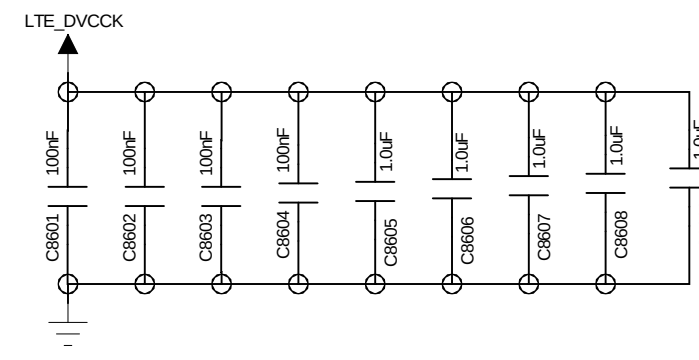
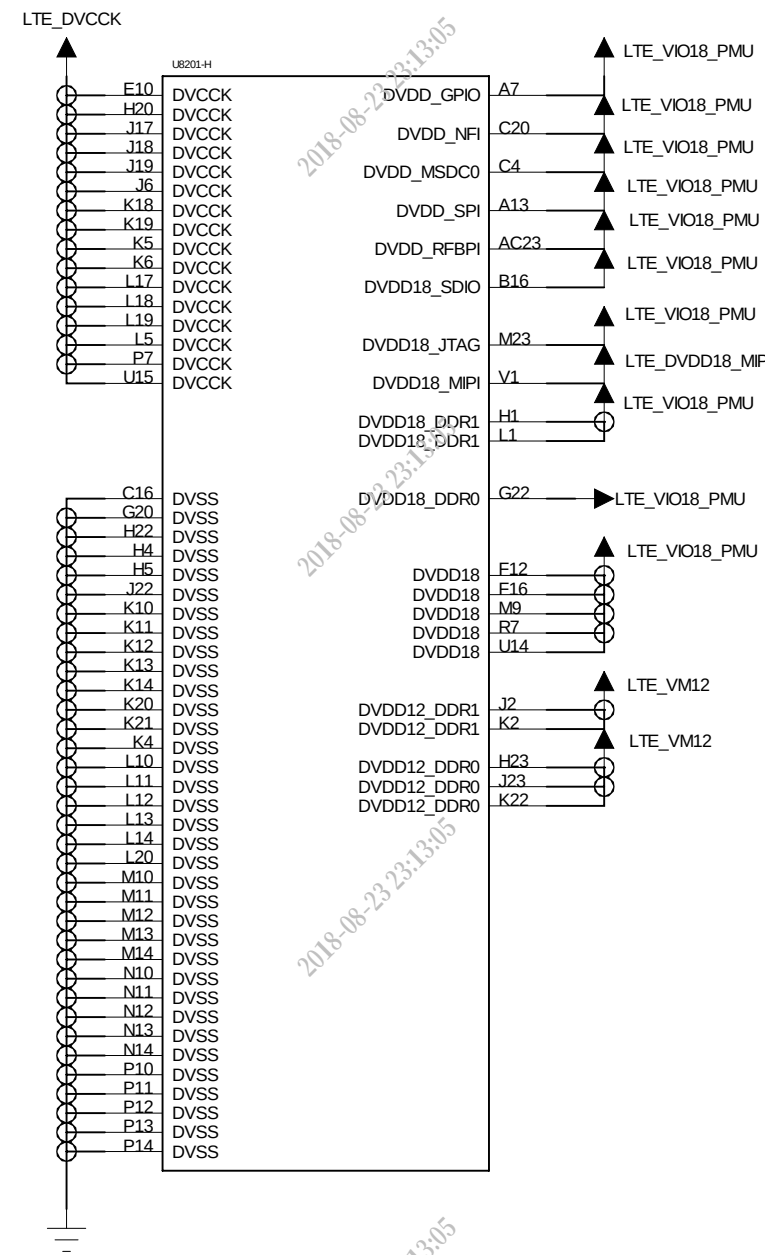
Gyroscope



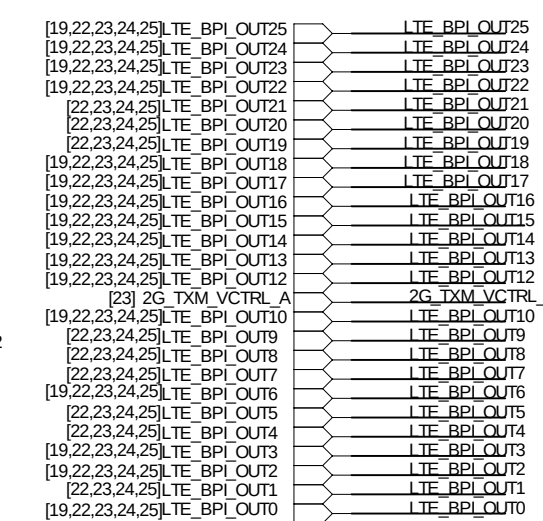
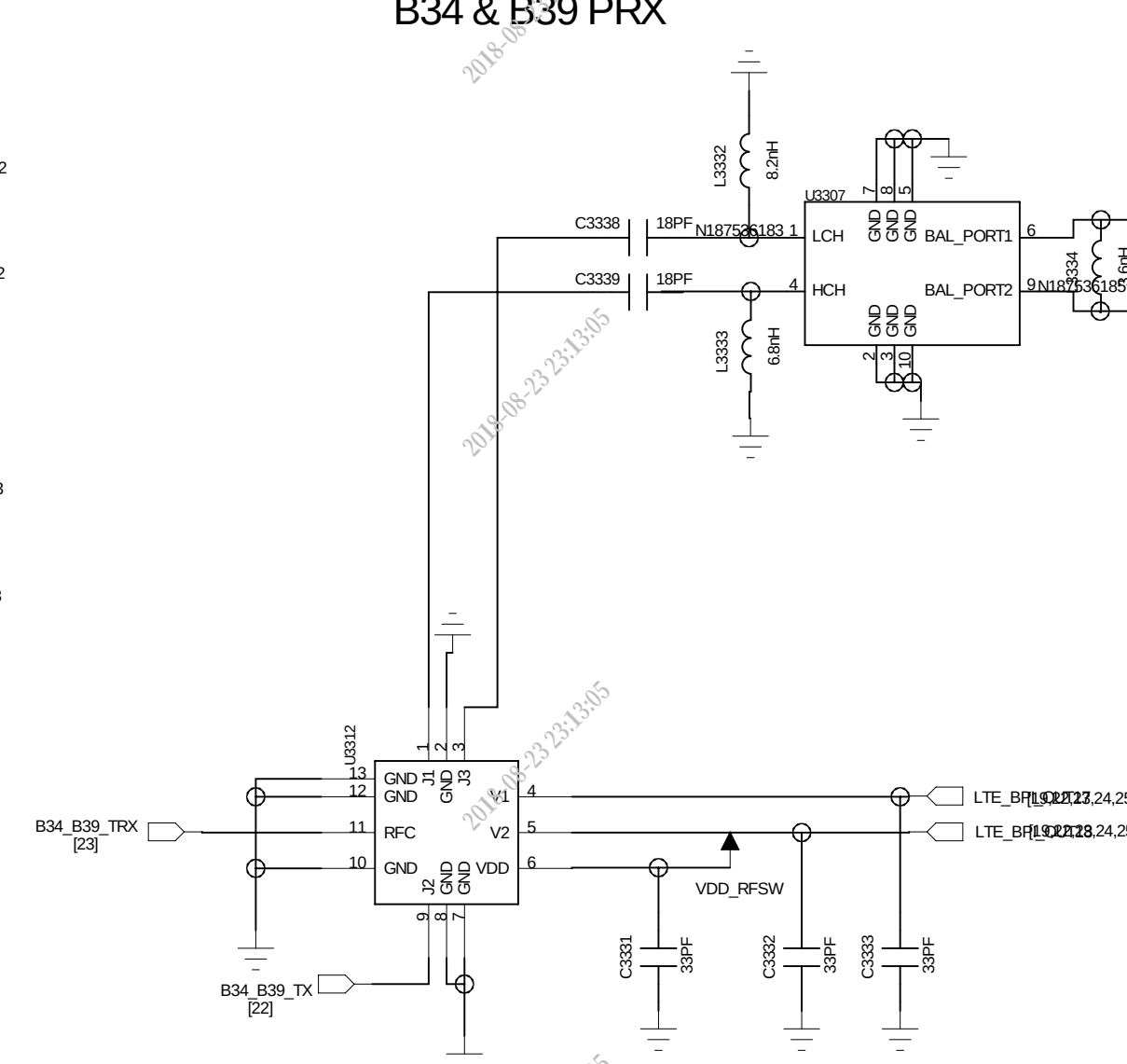
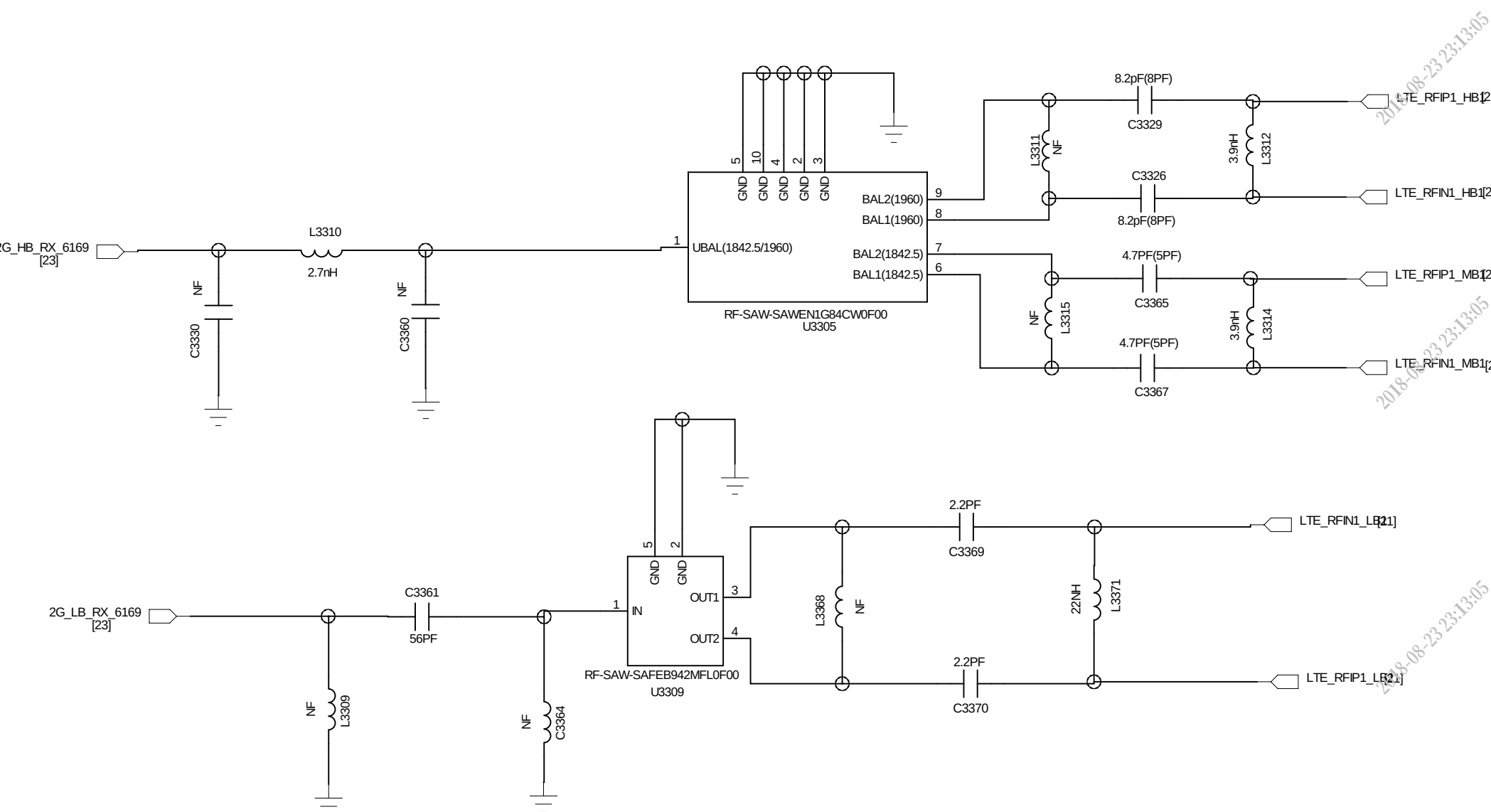
AD0=0 7b1101000 0x68
AD0=1 7b1101001 0x69





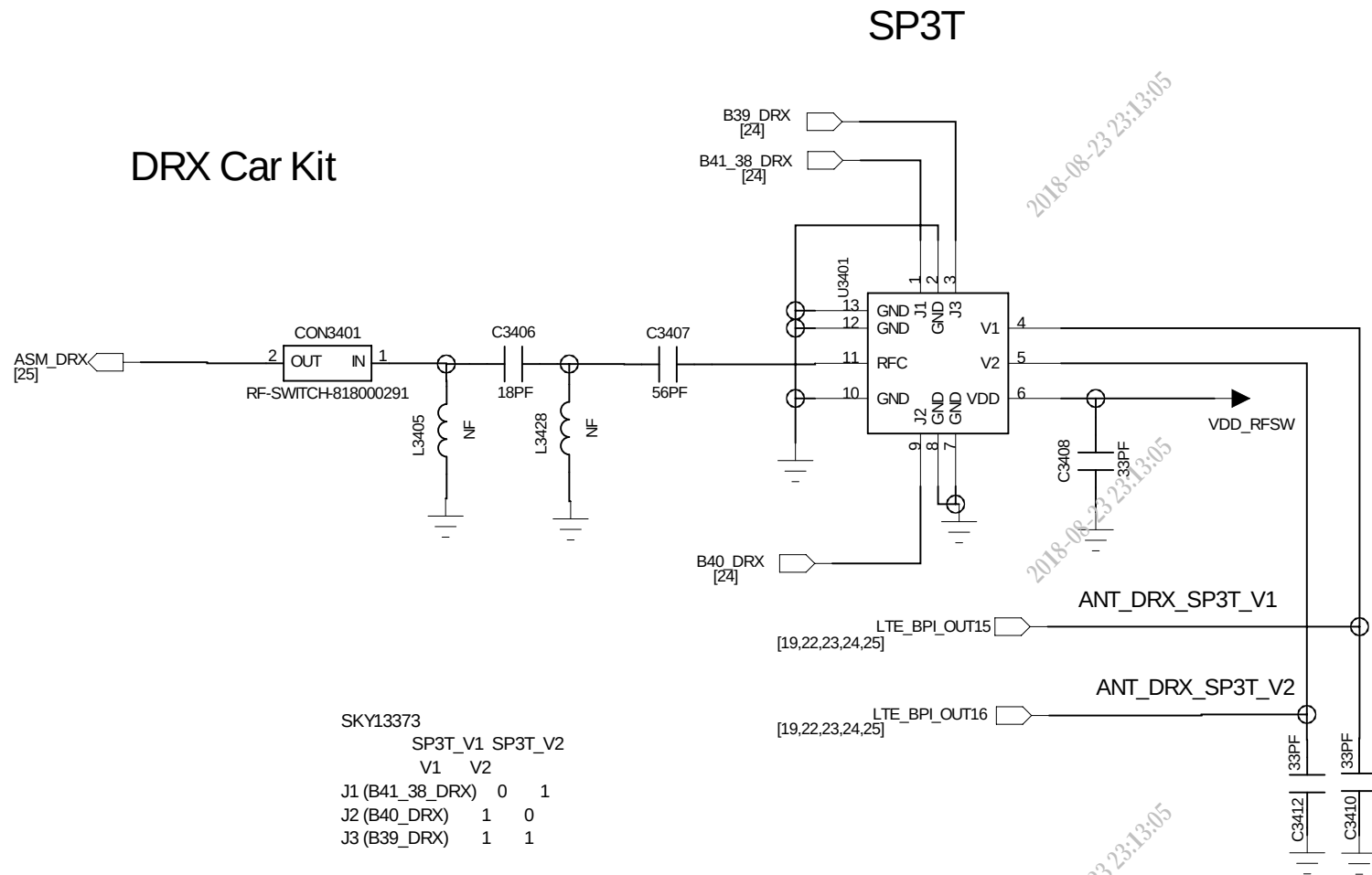




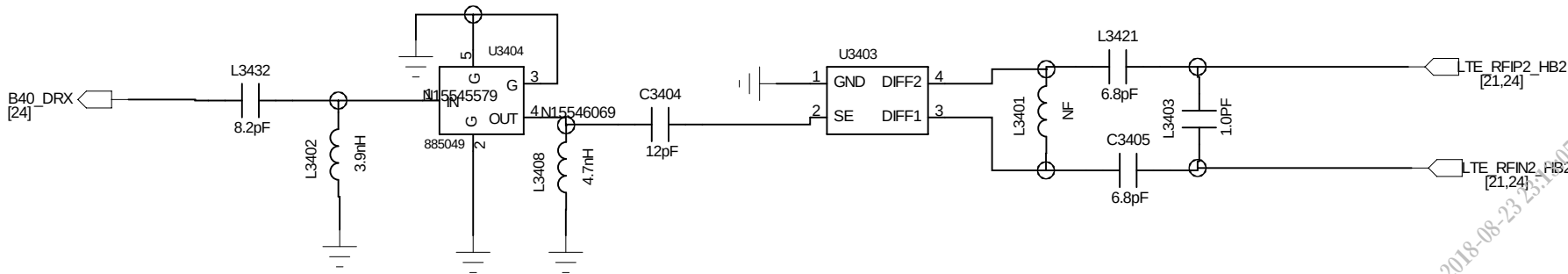


TITLE: <TITLE>		REV: <REV>
DOCUMENT NO.: 33_RF_MT6169_RF_PRX	SIZED: A1	
DEPARTMENT: Hardware DEPT.		
COMPANY: 		
DESIGNER: <DESIGNER>	Last Saved Date: 2014-7-3	SHEET: 23 OF 25

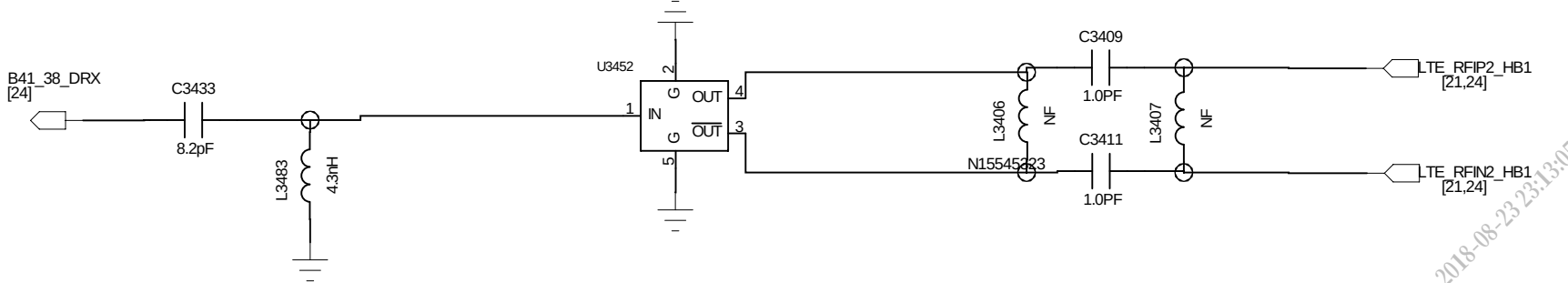
DRX ANT: 1880~2690MHz



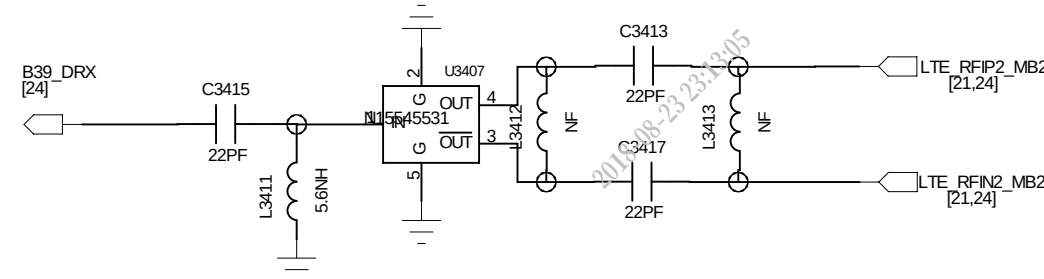
B40 DRX



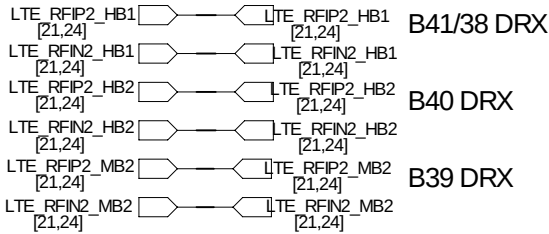
B41/38 DRX



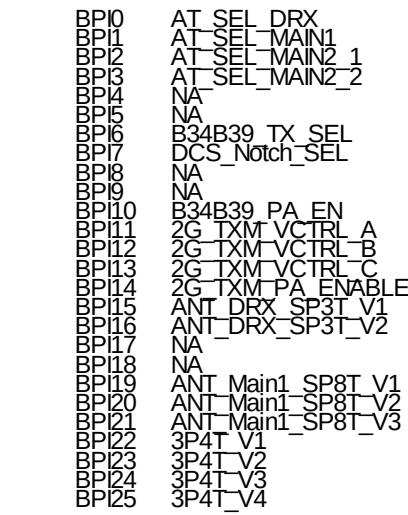
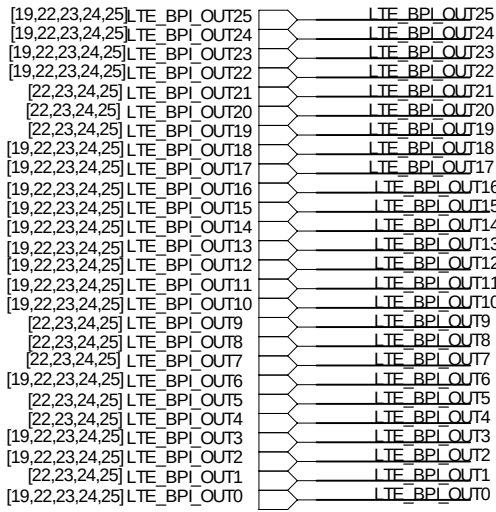
B39 DRX



DRX (To MT6169)



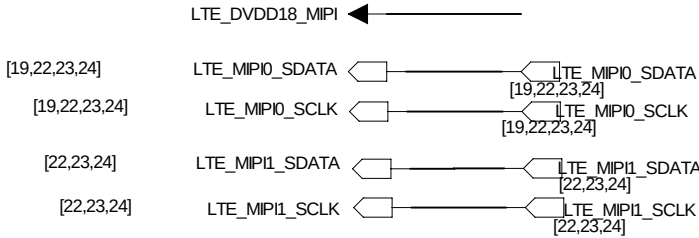
BPI config.



To LTE_VTCX01 (0ohm to MT6339)

VDD_RFSW

MIPI (To MT6290/6339)



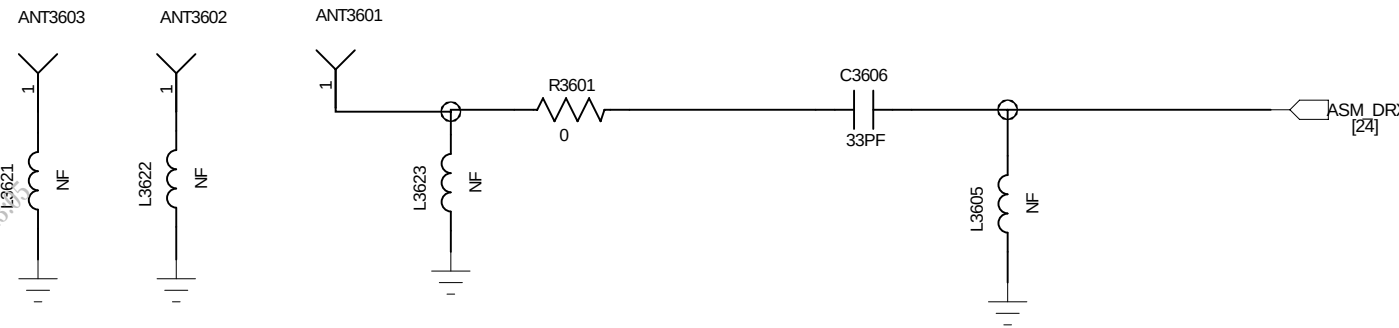
TITLE:	<TITLE>	REV:	<REV>
DOCUMENT NO:	34_RF_MT6169_RF_DRX	SIZED:	A1
DEPARTMENT:	Hardware DEPT.		
COMPANY:	WINGTEC		
DESIGNER:	<DESIGNER>	Last Saved Date:	2014-7-3
SHEET:	24	OF	25

2018-08-23 23:13:05

2018-08-23 23:13:05

2018-08-23 23:13:05

DRX ANT: 704~2700MHz



SKY13489
V1 Port
0 RF1
1 RF2

To LTE_VTCXO1 (Ohm to MT6339)
→ VDD_RF5W

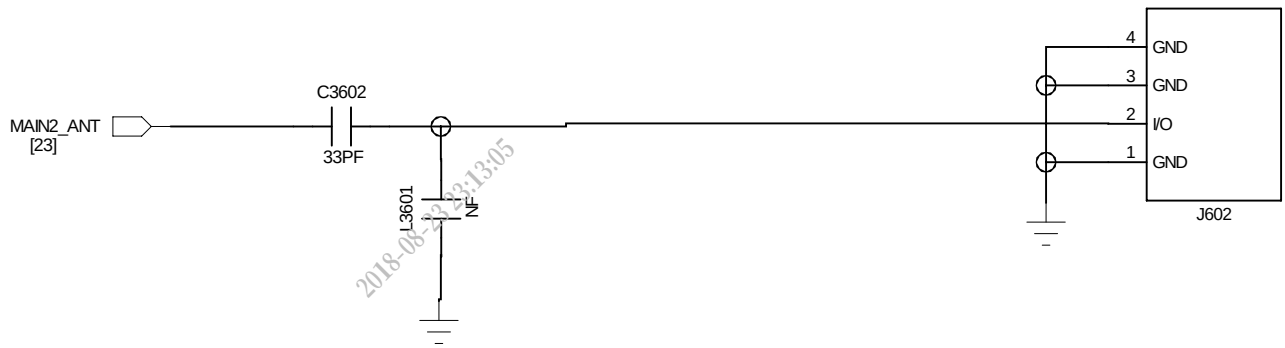
BPI config.
[19.22.23.24.25] LTE_BPI_OUT25
[19.22.23.24.25] LTE_BPI_OUT24
[19.22.23.24.25] LTE_BPI_OUT23
[19.22.23.24.25] LTE_BPI_OUT22
[22.23.24.25] LTE_BPI_OUT21
[22.23.24.25] LTE_BPI_OUT20
[22.23.24.25] LTE_BPI_OUT19
[19.22.23.24.25] LTE_BPI_OUT18
[19.22.23.24.25] LTE_BPI_OUT17
[19.22.23.24.25] LTE_BPI_OUT16
[19.22.23.24.25] LTE_BPI_OUT15
[19.22.23.24.25] LTE_BPI_OUT14
[19.22.23.24.25] LTE_BPI_OUT13
[19.22.23.24.25] LTE_BPI_OUT12
[19.22.23.24.25] LTE_BPI_OUT11
[19.22.23.24.25] LTE_BPI_OUT10
[22.23.24.25] LTE_BPI_OUT9
[22.23.24.25] LTE_BPI_OUT8
[22.23.24.25] LTE_BPI_OUT7
[19.22.23.24.25] LTE_BPI_OUT6
[22.23.24.25] LTE_BPI_OUT5
[22.23.24.25] LTE_BPI_OUT4
[19.22.23.24.25] LTE_BPI_OUT3
[22.23.24.25] LTE_BPI_OUT2
[19.22.23.24.25] LTE_BPI_OUT1
[19.22.23.24.25] LTE_BPI_OUT0
BPI_0 AT_SEL_DRX
BPI_1 AT_SEL_MAIN1
BPI_2 AT_SEL_MAIN2
BPI_3 NA
BPI_4 NA
BPI_5 B39_TX_SEL
BPI_6 B39_Noise_SEL
BPI_7 NA
BPI_8 B39_PA_EN
BPI_9 B39_TX_VCTR_A
BPI_10 B39_TX_VCTR_B
BPI_11 B39_TX_VCTR_C
BPI_12 B39_TX_PA_ENABLE
BPI_13 B39_TX_PA_ENABLE
BPI_14 B39_TX_PA_ENABLE
BPI_15 ANT_DRX_SP3T_V1
BPI_16 ANT_DRX_SP3T_V2
BPI_17 NA
BPI_18 ANT_Main1_SP8T_V1
BPI_19 ANT_Main1_SP8T_V2
BPI_20 ANT_Main1_SP8T_V3
BPI_21 3P4T_V1
BPI_22 3P4T_V2
BPI_23 3P4T_V3
BPI_24 3P4T_V4

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2018-08-23 23:13:05

2018-08-23 23:13:05

Main2 ANT: 704~2170MHz



2018-08-23 23:13:05

2018-08-23 23:13:05

2018-08-23 23:13:05

2018-08-23 23:13:05

2018-08-23 23:13:05

2018-08-23 23:13:05

2018-08-23 23:13:05

TITLE:	<TITLE>	REV:	<REV>
DOCUMENT NO:	36_RF_AT	SIZED:	A1
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	<DESIGNER>	Last Saved Date:	2014-7-3
SHEET:	25	OF	25